

Prometheus

Create Agent Principal. Assertion Act. A consumer can check that Assertion Act. Decommission. Those four acts are the product.

21 August 2026 · Names locked 21 August 2026 · Prepared for Jason Gale · Laboratory side project. Not a company product.

1. The claim in one page

A classical identity provider does five later steps. A directory creates a name. A login proves a human. Groups or roles grant authorization after that login. A review governs that grant. A vault holds a standing secret.

That order assumes a person who can attest intent. A software agent cannot do that. The grant arrives too late. The secret stays too long. A name lookup is not proof of one act.

Prometheus inverts the order. Create Agent Principal is the privilege. Assertion Act is a document of one act. Decommission is portable. A child is a narrower act, not a role in a catalog.

The kernel is testable. This brief does not say the bet is won. Human identity replacement is not this laboratory.

2. Why classical identity fails

Classical identity was built for a human. The human has a name in a directory. The human proves that name with a login. After login, a group or a role becomes the grant. A later review asks whether that grant is still right. A vault keeps a password or a key so the next login can happen.

Each step is later than Create Agent Principal. The system first creates an empty principal. Intent is attached afterwards. Decommission is a disable flag in the same directory, or an expiry on a

certificate. A second party that is not a replica of that directory cannot be sure the principal is decommissioned.

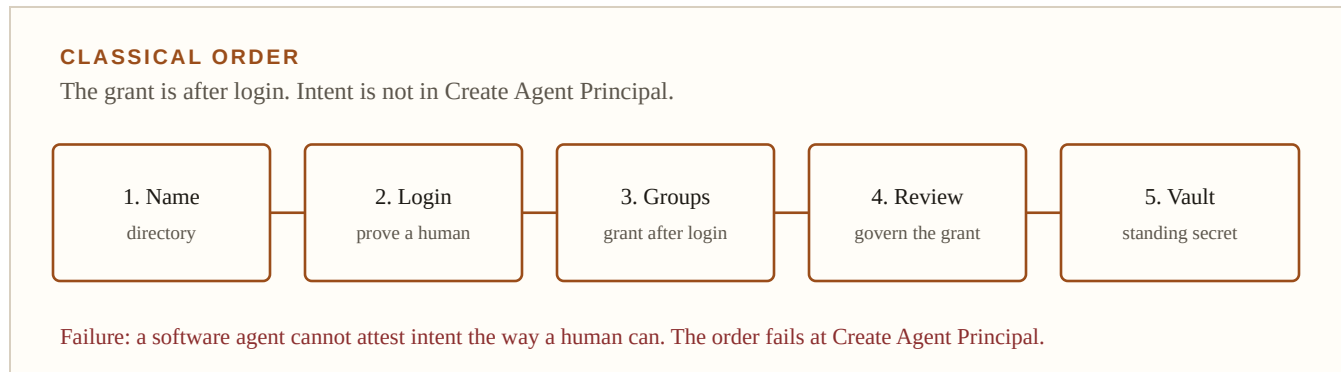


Figure 1. Classical identity is five later steps. The privilege is not Create Agent Principal.

Three holes follow.

1. Authorization is a group after login. It is not a capability written at Create Agent Principal.
2. A standing secret outlives the act. The next caller can reuse it.
3. Decommission lives in the issuing directory. A verifier that does not copy that directory cannot refuse a later Assertion Act.

3. Why an ecosystem grows around that stack

The five later steps are incomplete on their own. Each hole grows a later product.

The directory is empty of intent, so a connector copies groups into every application. The grant is wide, so an access-review product asks a human to recertify. The standing secret is dangerous, so a secrets broker rotates it. The workload still needs a name on the wire, so a mesh mints a short-lived certificate for a name. The certificate expiry is then treated as if it were Decommission. It is not Decommission. A new certificate can be minted for the same name.

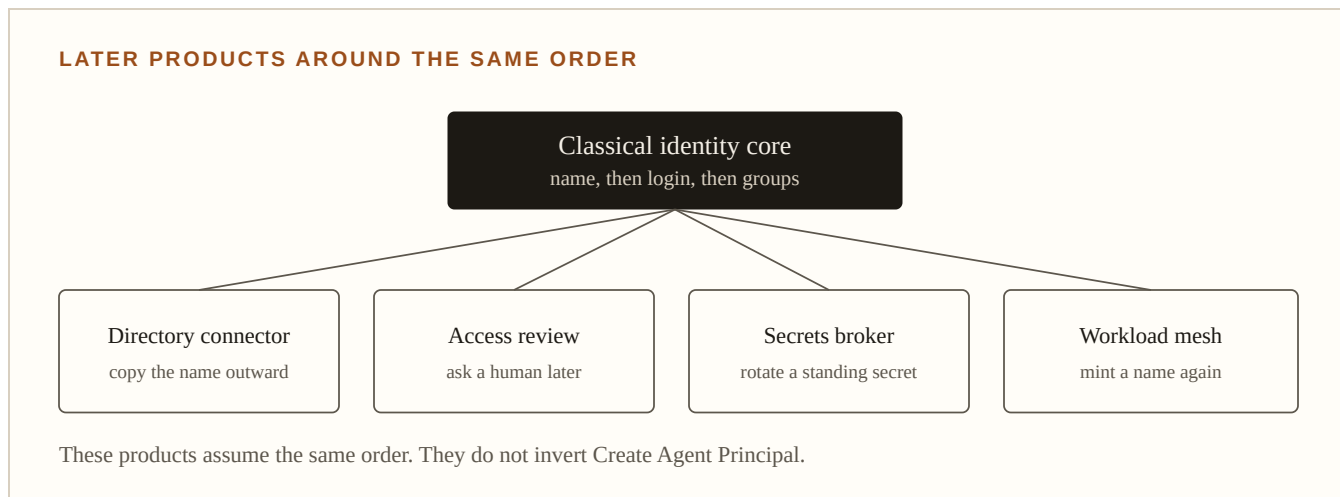


Figure 2. Later products grow because the core never wrote intent, Decommission, and an Assertion Act at Create Agent Principal.

That ecosystem is rational for humans. It is the wrong factory for software agents. An agent does not recertify a group. An agent does not rotate a password. An agent that receives a mesh name still has no portable Decommission and no intent at Create Agent Principal.

4. Why a software agent needs a different order

A software agent acts without a person in the path who can attest intent. If Create Agent Principal is only a name, the agent can later obtain a wide group, keep a standing secret, and keep acting after a human believes it was disabled.

The agent needs four properties that the classical stack does not give at Create Agent Principal.

1. The privilege is Create Agent Principal. Intent and destination are set then. The instance cannot raise its own authorization limit.
2. Each act is an Assertion Act document. A consumer verifies that document. The consumer does not look up a name and enumerate groups.
3. Decommission can travel to a verifier that does not copy the identity store.
4. A child is a narrower act. A child cannot exceed its parent. Spawn is not a role catalog.

Records persist. Bearer credentials do not. The holder key proves possession for one challenge. The holder key is not the instance name. A short certificate life is not Decommission.

5. What Prometheus inverts

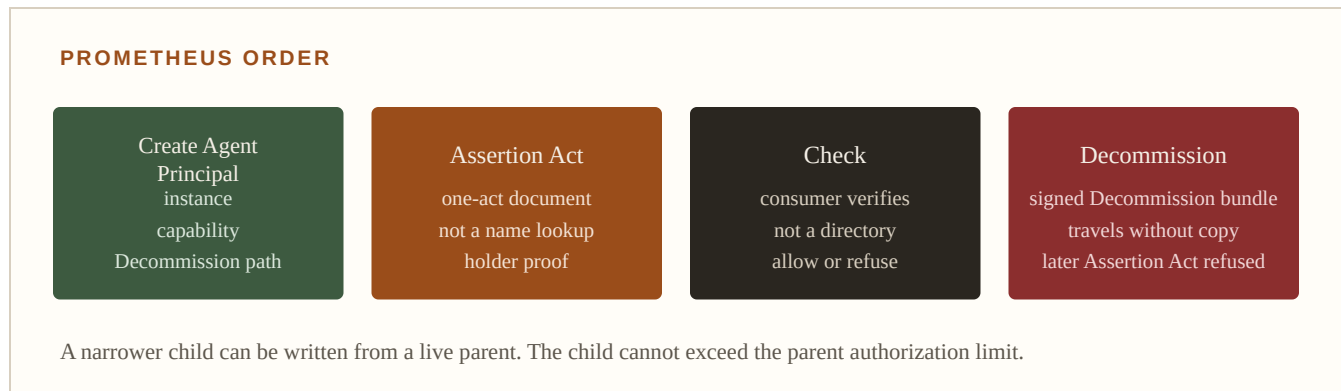


Figure 3. The product is four acts. Public key certificates are a presentation on-ramp. They are not the identity record.

Five identity records exist: agent type, instance, capability, chain, and issuer. There is no sixth identity record. Five kernel system calls exist: mint, verify, attenuate, present, and kill. Create Agent Principal and spawn sit on mint. The laboratory path names stay mapped. Check sits on verify.

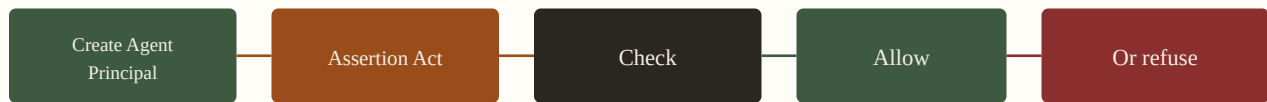
The identity root is Module-Lattice Digital Signature Algorithm. A new Create Agent Principal write refuses a classical-only root. A laboratory capability envelope may travel on the wire. That envelope is not the identity record.

6. Authorization and Decommission flows

One-store act

An operator runs Create Agent Principal for an agent type. The write sets intent and destination. The agent produces an Assertion Act and proves holder possession for one challenge. A check host verifies the document and allows or refuses the tool action. The check host is not a directory. After Decommission, a historical Assertion Act is refused. Unknown does not mean live. There is no force-allow.

ONE STORE · AUTHORIZATION FLOW



After Decommission, the same historical Assertion Act is refused. Short certificate life is not Decommission.

Holder proof is a signature over one challenge. The holder key is not the instance name.

A child spawn must stay inside the parent intent and destination.

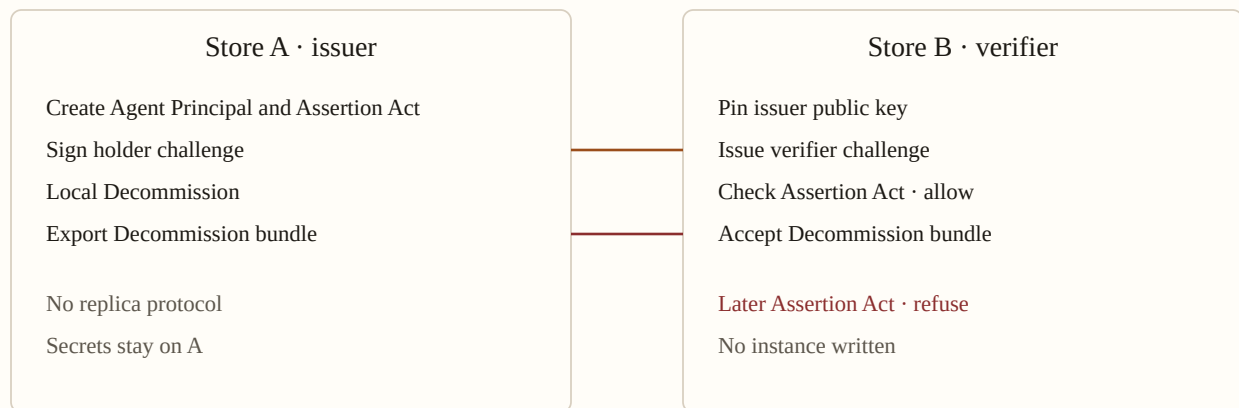
Figure 4. Check sits on verify. The host allows or refuses one Assertion Act. Decommission ends the principal.

Two-store portable Decommission

Store A issues. Store B verifies. Store B does not mint. Store B does not receive issuer secrets.

Store B pins the issuer public key. Store B issues a verifier challenge. Store A signs that challenge with the holder key. Store B checks the Assertion Act with the holder signature. An honest Assertion Act allows. Store A exports a signed Decommission bundle. Store B accepts that bundle and writes no instance record. A later check of the same historical Assertion Act is refused. Decommission traveled. The identity store was not copied.

TWO STORES · PORTABLE DECOMMISSION



Issuer seal and previous-key kill travel the same way: public artifacts, verifier state on the issuer record, no sixth identity record.

Figure 5. A verifier that is not a replica can refuse after Decommission. That is portable Decommission.

Agent runtime before a tool

A laboratory runtime follows one well-known check document on the loopback address. The runtime requests a verifier challenge, posts the documented check with a caller-supplied holder signature, and exits allowed or refused. The runtime does not read holder secrets. The runtime is

not an issuer. The runtime is not a directory. After Decommission accept, the same verb exits refused.

7. How this closes the gaps

CLASSICAL HOLE	PROMETHEUS CLOSE
Create Agent Principal is only a name. Intent arrives later as a group.	One Create Agent Principal write holds the instance, the first capability, the holder key, and the Decommission path. Authorization limit is set then.
A consumer looks up a name and enumerates groups.	A consumer verifies an Assertion Act. The consumer is a check, not a directory.
A standing secret outlives the act.	Records persist. Bearer credentials do not. Holder proof is one challenge.
Decommission lives only in the issuing directory.	A signed Decommission bundle travels. A verifier accepts it and writes no instance record.
Certificate expiry is treated as Decommission.	Short certificate life is not Decommission. Decommission ends the principal.
A child is a role copy.	Spawn writes a narrower child. The child cannot exceed the parent.
A later mesh mints a name as identity.	A public key certificate is an on-ramp. The instance name must not become a distinguished name.
Unknown is treated as live.	Unknown does not mean live. Enforcement fails closed. There is no force-allow.

8. What is built now

The kernel on one store can init, add an agent type, Create Agent Principal, challenge, Assertion Act, check, spawn, and Decommission. A second store can accept an issuer public key, accept an act bundle, and accept a Decommission bundle without minting.

A check host binds the loopback address only. A person can test Create Agent Principal, Assertion Act, check, and Decommission in a browser on this computer. A laboratory runtime can complete one check before a tool and exit allowed or refused. Two stores can prove portable Decommission. Issuer seal and previous-key kill travel as verifier state on the issuer record.

A visible walk answered five questions yes. Create Agent Principal writes instance, capability, and Decommission in one persist. A second party can verify an act without copying the identity store. Decommission can travel and refuse a later Assertion Act. A runtime can consume an Assertion Act without becoming a directory. A stolen or resigned wrap fails closed after Decommission.

This host is not a public listener. A public listener waits for a real host name. Init of a store still uses the command line. A stolen issuer secret can still mint an Assertion Act that a verifier allows until that verifier accepts issuer seal or a previous-key kill. Those limits are stated. They are not hidden.

9. What this is not

Prometheus is not a human identity provider. Prometheus is not a secrets-management product. Prometheus is not a public directory. Prometheus is not a company product. Findings here are not a later identity platform until a real host name exists and a runtime will not call loopback check.

The kernel is testable. This page does not say the bet is won. Jason Gale decides whether this agent identity is worth pursuing.

Source of truth: Prometheus laboratory · 21 August 2026. Simplified Technical English. Full technical words. No standing secret is copied between stores.